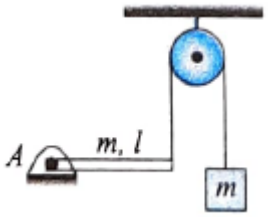
# Quizrr Chapter-Wise Test for JEE Main - 2025

**Q.1**

Uniform rod $AB$ is hinged at end $A$ in horizontal position as shown in the figure. The other end is connected to a block through a massless string as shown. The pulley is smooth and massless. Masses of block and rod is same and is equal to $m$. Then acceleration of block just after release from this position is




**(A)** $6g/13$

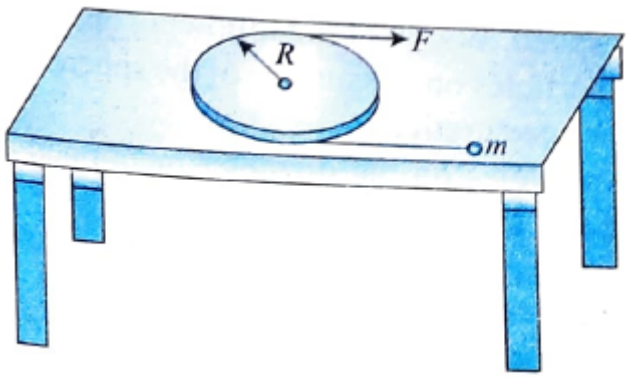
**(B)** $g/4$

**(C)** $3\ g/8$

**(D)** None

**Q.2**

A disc of mass $m$ and radius $R$ placed on a smooth horizontal table as shown in figure. A light ideal string passes through the disc such that the one end of the string is attached to a small body of mass $m$ and the other end is being pulled with a force $F$. The circumference of the disc is sufficiently rough so that the string does not slip over it. Find acceleration of the small body.




**(A)** $\frac{F}{4m}$

**(B)** $\frac{F}{m}$

**(C)** $\frac{F}{2m}$

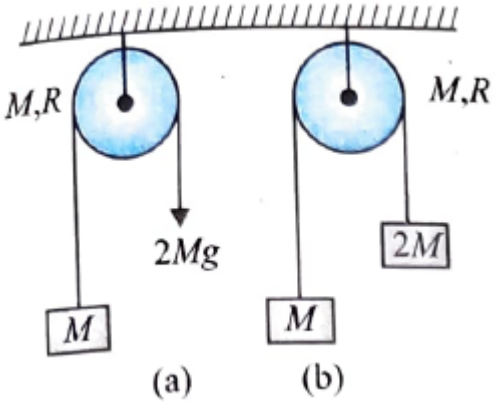
**(D)** $\frac{F}{3m}$

**Q.3**

A cord is wrapped on a pulley (disk) of mass $M$ and radius $R$ as shown in figure. To one end of the cord, a block of

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mass M is connected as shown and to other end in (a) a force of 2 Mg and in (b) a block of mass $2M$. Let angular acceleration of the disk in $A$ and $B$ is $\alpha_A$ and $\alpha_B$, respectively, then (cord is not slipping on the pulley).




**(A)** $\alpha_A = \alpha_B$

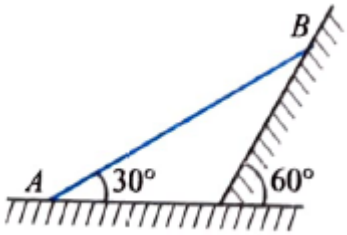
**(B)** $\alpha_A > \alpha_B$

**(C)** $\alpha_A < \alpha_B$

**(D)** None of these

**Q.4**

In the figure shown, the instantaneous speed of end $A$ of the rod is $v$ to the left. The angular velocity of the rod of length $L$ must be




**(A)** $\frac{v}{2L}$

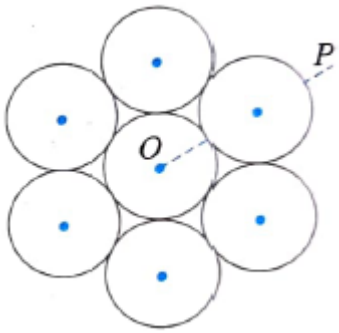
**(B)** $\frac{v}{L}$

**(C)** $\frac{v\sqrt{3}}{2L}$

**(D)** none of these

**Q.5**

Seven identical circular planar disks, each of mass $M$ and radius $R$ are welded symmetrically as shown. The moment of inertia of the arrangement about the axis normal to the plane and passing through the point $P$ is




**(A)** $\frac{181}{2}MR^2$

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**(B)** $\frac{19}{2}MR^2$

**(C)** $\frac{55}{2}MR^2$

**(D)** $\frac{73}{2}MR^2$

**Q.6**

A string is wrapped around a cylinder of mass $M$ and radius R. The string is pulled vertically upwards to prevent centre of mass from falling as the cylinder upwind string. The tension in the string is

**(A)** $2M\ \mathrm{g}/3$

**(B)** $\mathrm{Mg}/2$

**(C)** $\mathrm{Mg}/3$

**(D)** $M\ \mathrm{g}/6$

**Q.7**

A pulley of radius $2\ \mathrm{m}$ is rotated about its axis by a force $F = \left(20t - 5t^2\right)$ newton (where $t$ is measured in seconds) applied tangentially. If the moment of inertia of the pulley about its axis of rotation is $10\ \mathrm{kg\ m^2}$, the number of rotations made by the pulley before its direction of motion is reversed, is

**(A)** less than 3

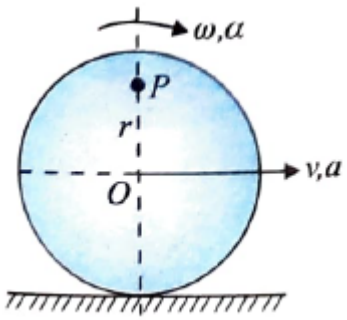
**(B)** more than 3 but less than 6

**(C)** more than 6 but less than 9

**(D)** more than 9

**Q.8**

A disc of radius $R$ rolls on a horizontal ground with linear acceleration $a$ and angular acceleration $\alpha$ as shown in figure. The magnitude of acceleration of point $P$ as shown in the figure at an instant when its linear velocity is $v$ and angular velocity is $\omega$ will be




**(A)** $\sqrt{(a+r\alpha)^2+\left(r\omega^2\right)^2}$

**(B)** $\frac{\mathrm{ar}}{R}$

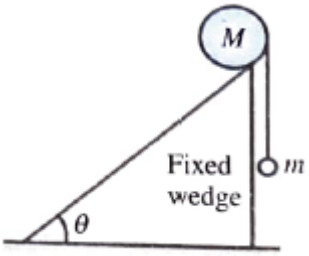
**(C)** $\sqrt{r^2\alpha^2+r^2\omega^4}$

**(D)** $r\alpha$

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**Q.9**

A uniform cylinder of mass $M$ lies on a fixed plane inclined at an angle $\theta$ with horizontal. A light string is tied to the cylinder at the right most point, and a mass $m$ hangs from the string, as shown. Assume that the coefficient of friction between the cylinder and the incline plane is sufficiently large to prevent slipping. For the cylinder the remain static, the value of $m$ is




**(A)** $\frac{M \sin\theta}{1-\sin\theta}$

**(B)** $\frac{M \cos\theta}{1+\sin\theta}$

**(C)** $\frac{M \sin\theta}{1+\sin\theta}$

**(D)** $\frac{M \cos\theta}{1-\sin\theta}$

**Q.10**

A uniform wooden stick of mass $1.6\ \text{kg}$ and length $l$ rest in an inclined manner on a smooth, vertical wall of height $h(<l)$ such that a small portion of the stick extends beyond the wall. The reaction force of the wall on the stick is perpendicular to the stick. The stick makes an angle of $30^\circ$ with the wall and the bottom of the stick is on a rough floor. The reaction of the wall on the stick is equal in magnitude? to the reaction of the floor on the stick. The ratio $h/l$ and the frictional force $f$ at the bottom of the stick are $(g = 10\ \text{ms}^\circ)$

**(A)** $\frac{h}{l} = \frac{\sqrt{3}}{16}, f = \frac{16\sqrt{3}}{3}\ \text{N}$

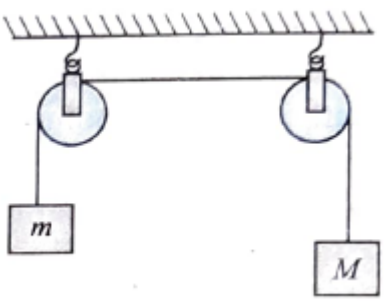
**(B)** $\frac{h}{l} = \frac{3}{16}, f = \frac{16\sqrt{3}}{3}\ \text{N}$

**(C)** $\frac{h}{l} = \frac{3\sqrt{3}}{16}, f = \frac{8\sqrt{3}}{3}\ \text{N}$

**(D)** $\frac{h}{l} = \frac{3\sqrt{3}}{16}, f = \frac{16\sqrt{3}}{3}\ \text{N}$

**Q.11**

Each pulley in figure has radius $r$ and moment of inertia_ The acceleration of the block is




**(A)** $\frac{(M-m)g}{\left(M+m+\frac{2I}{r^2}\right)}$

**(B)** $\frac{(M-m)g}{\left(M+m-\frac{2I}{r^2}\right)}$

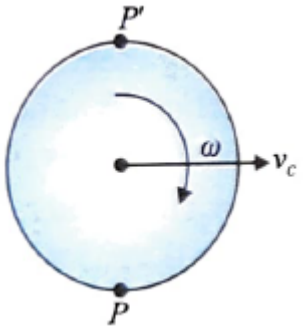
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**(C)** $\frac{(M-m)g}{\left(M+m+\frac{I}{r^2}\right)}$

**(D)** $\frac{(M-m)g}{\left(M+m-\frac{I}{r^2}\right)}$

**Q.12**

A sphere is moving towards the positive $x$-axis with a velocity $v_c$ and rotates clockwise with angular speed $\omega$ as shown in figure, such that $v_c > \omega R$. The instantaneous axis of rotation will be




**(A)** on point $P$

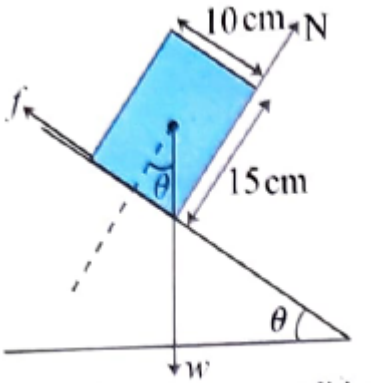
**(B)** on point $P'$

**(C)** inside the sphere

**(D)** outside the sphere

**Q.13**

A block of base $10\ \text{cm} \times 10\ \text{cm}$ and height $15\ \text{cm}$ is kept on an inclined plane. The coefficient of friction between them is 3 . The inclination $\theta$ of this inclined plane from the horizontal plane is gradually increased from $0^\circ$. Then



FBD at just toppling condition

**(A)** at $\theta = 30^\circ$, the block will start sliding down the plane

**(B)** the block will remain at rest on the plane up to certain $\theta$ and then it will topple

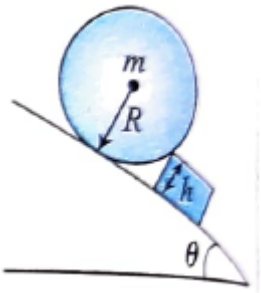
**(C)** at $\theta = 60^\circ$, the block will start sliding down the plane and continue to do so at higher angles

**(D)** at $\theta = 60^\circ$, the block will start sliding down the plane and on further increasing $\theta$, it will topple at certain $\theta$

**Q.14**

Find the minimum height of the obstacle so that the sphere. can stay in equilibrium.

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**(A)** $\frac{R}{1+\cos\theta}$

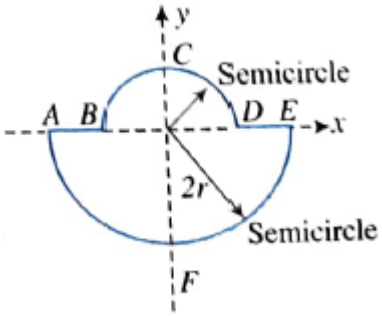
**(B)** $\frac{R}{1+\sin\theta}$

**(C)** $R(1-\sin\theta)$

**(D)** $R(1-\cos\theta)$

**Q.15**

A uniform thin rod is bent in the form of closed loop $ABCDEFA$ as shown in the figure. The ratio of moment of inertia of the loop about $x$-axis to that about $y$-axis is




**(A)** $>1$

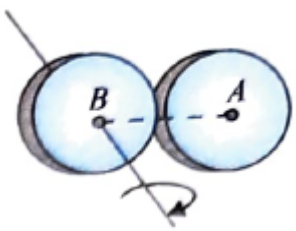
**(B)** $<1$

**(C)** $=1$

**(D)** $=1/2$

**Q.16**

Two thin discs, each of mass $M$ and radius $r$ metre, are attached as shown in figure, to form a rigid body. The rotational inertia of this body about an axis perpendicular to the plane of disc $B$ passing through its centre is




**(A)** $2Mr^2$

**(B)** $3\mathrm{Mr}^2$

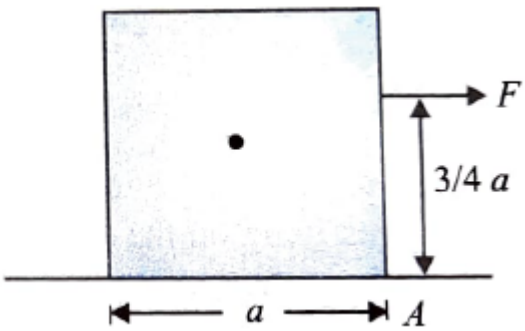
**(C)** $4\mathrm{Mr}^2$

**(D)** $5\mathrm{Mr}^2$

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**Q.17**

A cube of side $a$ and mass $m$ is to be tilted at point $A$ by applying a force $F$ as shown in figure. The minimum force required is




**(A)** $\mathrm{mg}$

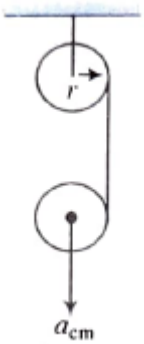
**(B)** $\frac{2}{3}mg$

**(C)** $\frac{3}{2}mg$

**(D)** $\frac{3}{4}mg$

**Q.18**

Two identical uniform discs of mass $m$ and radius $r$ are arranged as shown in the figure. If $\alpha$ is the angular acceleration of the lower disc and $a_{\mathrm{cm}}$ is acceleration of centre of mass of the lower disc, then relation among $a_{\mathrm{cm}}$, $\alpha$ and $r$ is




**(A)** $a_{\mathrm{cm}} = \frac{\alpha}{r}$

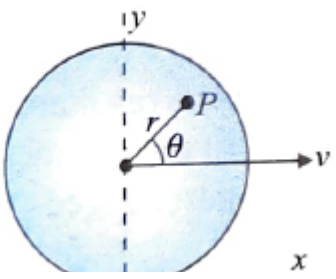
**(B)** $a_{\mathrm{cm}} = 2\alpha r$

**(C)** $a_{\mathrm{cm}} = \alpha r$

**(D)** none of these

**Q.19**

A disc of radius $R$ rolls without slipping at speed $v$ along positive $x$-axis. Velocity of point $P$ at the instant shown in figure is




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**(A)** $\vec{v}_p = \left(v + \frac{vr\sin\theta}{R}\right)\hat{i} + \frac{vr\cos\theta}{R}\hat{j}$

**(B)** $\vec{v}_{\bar{F}} = \left(v + \frac{vr\sin\theta}{R}\right)\hat{i} - \frac{vr\cos\theta}{R}\hat{j}$

**(C)** $\vec{v}_F = \frac{vr\sin\theta}{R}\hat{i} + \frac{vr\cos\theta}{R}\hat{j}$

**(D)** $\vec{v}_f = \frac{\nu r\sin\theta}{R}\hat{i} - \frac{vr\cos\theta}{R}\hat{j}$

**Q.20**

The wheels of an airplane are set into rotation just before landing so that the wheels do not slip on the ground. If the airplane is travelling in the east direction, what should be the direction of angular velocity vector of the wheels?

**(A)** East

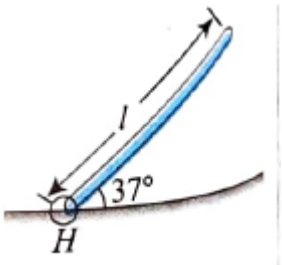
**(B)** West

**(C)** South

**(D)** North

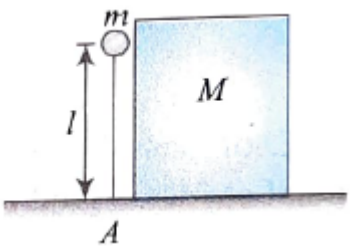
**Q.21**

A uniform rod of mass $m = 2\sqrt{10}\ \text{kg}$ and length $l$ can rotate in vertical plane about a smooth horizontal axis hinged at point $H$. Angular acceleration $a$ of the rod just after it released from initial position making an angle of $37^\circ$ with horizontal from rest. Find force (in $\text{N}$) exerted by the hinge just after the rod is released from rest.




**Q.22**

A weightless rod of length $l$ with a small load of mass m at the end is hinged at point $A$ as shown and occupies a strictly vertical position, touching a block of mass $M$. A light jerk sets the system in motion. For what mass ratio $\text{M/m}$ will the rod form an angle $\theta = \pi/6$ with the horizontal at the momentum of the separation from the block?

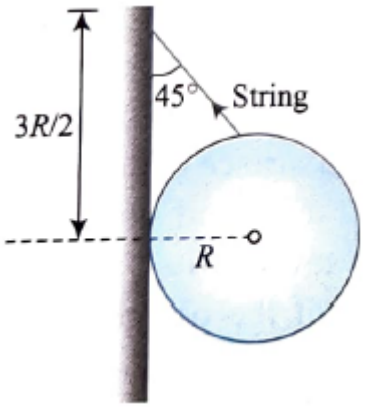



**Q.23**

A sphere of radius $R$ is supported by a rope attached to the wall. The rope makes an angle $\theta = 45^\circ$ with respect to the wall. The point where the rope is attached to the wall is at a distance of $3R/2$ from the point where the sphere

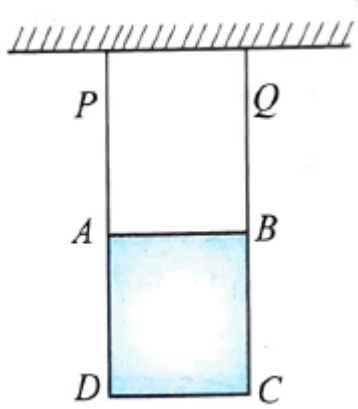
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touches the wall. Find the minimum coefficient of friction $(\mu)$ between the wall and the sphere for this equilibrium to be possible. Use the intregral part only.



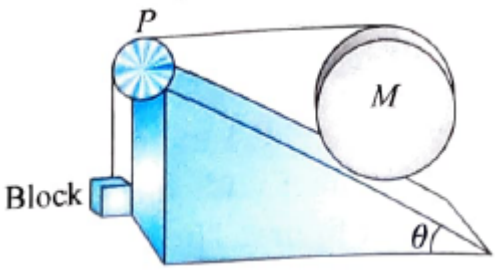

**Q.24**

A square plate $ABCD$ of mass $m$ and side $l$ is suspended with the help of two ideal strings P and Q as shown. Determine the acceleration $(\text{in m/s}^2)$ of corner $A$ of the square just at the moment the string $Q$ is cut. $(g = 10\ \text{m/s}^2)$



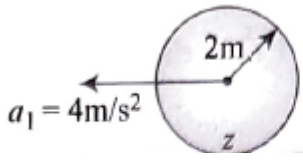

**Q.25**

In the arrangement shown in figure the cylinder of mass $M = 9\ \text{kg}$ is at rest on an incline. The plane is inclined at angle $37^\circ$ with horizontal. The string between the cylinder and the pulley $(P)$ is horizontal. Find the minimum the mass of the block (in $\text{kg}$) to keep the system in equilibrium.




**Q.26**

In figure, a sphere of radius $2\ \text{m}$ rolls on a plank. The accelerations of the sphere and the plank are indicated. Find the value of $\alpha$ in $\text{rad/s}^2$.

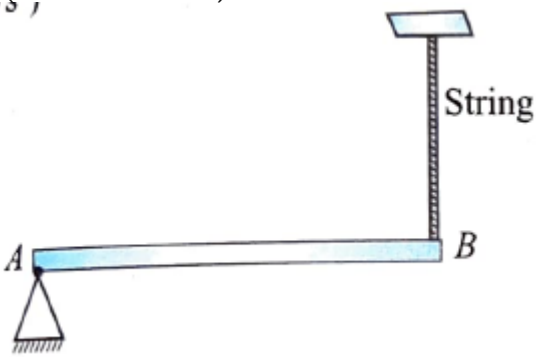



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**Q.27**

A uniform rod $AB$ of mass $2\ \text{kg}$ is hinged at one end $A$. The rod is kept in the horizontal position by a massless string tied to point $B$. Find the reaction of the hinge (in $\text{N}$) on end $A$ of the rod at the instant when string is cut. $(g = 10\ \text{m/s}^2)$



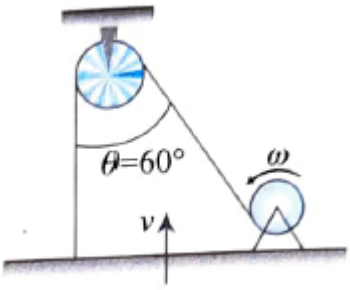

**Q.28**

Auniform cylinder rests on a cart as shown. The coefficient of static friction between the cylinder and the cart is $0.5$ . If the cylinder is $4\ \text{cm}$ in diameter and $10\ \text{cm}$ in height, then what is the minimum acceleration (in $\text{ms}^{-2}$) of the cart 'needed to cause the cylinder to tip over?



**Q.29**

In the instant shown in the figure, the board is moving vertically with constant velocity $(v)$. The drum wind up at a constant rate $\omega = 15\text{rad/s}$. If the radius of the drum is $R = 10\ \text{cm}$ and the board always remains horizontal. Find the velocity of board at this moment( in $\text{m/s}$).




**Q.30**

A wheel of radius R=1 m is rolling without sliding uniformly on a horizontal surface. Find the radius of curvature of the path (in m), of a point on its circumference when it is at highest point in its path?

Answers & Solutions

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**Q.1 Answer:**

$3\ g/8$

**Solution:**

**As shown in figure, $T$ = Tension of string, $a$ = acceleration of block, $N$ = Normal reaction at hinged point.**

**From the FBD of a block,**

$mg - T = ma \quad \ldots\ldots\ldots\ldots 1$

**From FBD of rod,**

**Taking Moments about hinged point,** $T \times 1 - mg \times \frac{1}{2} = \frac{ml^2}{3} \times \alpha$

$\Rightarrow T - mg = \frac{ml}{3} \times a$

**Using constraint relation** $a = a \times 1 \quad \ldots\ldots\ldots 2$

$\Rightarrow T - mg = \frac{ma}{3} \ldots\ldots\ldots\ldots 3$

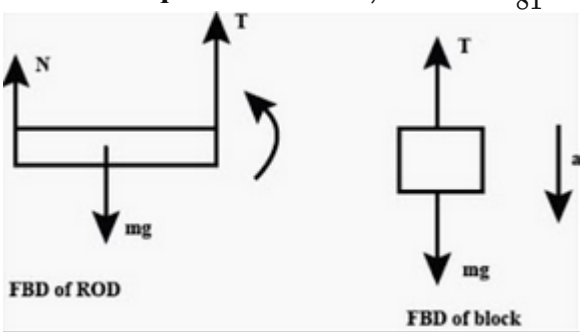
**Solving Equation 1 and 2 ,**

$\Rightarrow a = \frac{3\,g}{8}, \ldots\ldots\ldots 4$

**Using equation 4 in equation 1 ,**

$\Rightarrow T = \frac{5mg}{8}$

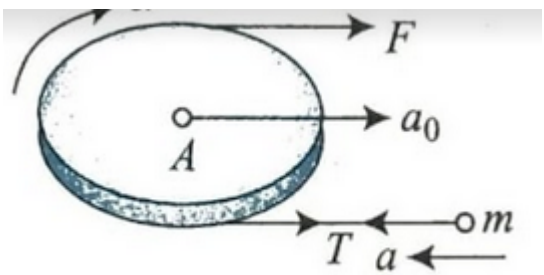
**From the equation 2 and 4,** $\Rightarrow a = \frac{3g}{81}$




**Q.2 Answer:**

$\frac{F}{4m}$

**Solution:**




$\Rightarrow R\alpha = a_0 + a$ **...(i)**

**Other equations are**

$T = ma \quad \ldots(\text{ii})$

$F + T = ma_0 \quad \ldots(\text{iii})$

and $(F - T)R = \frac{1}{2}mR^2 \cdot \alpha$

$\Rightarrow \quad F - T = \frac{1}{2}mR\alpha \quad \ldots(\text{iv})$

**Using (i) in (iv)** $F - T = \frac{1}{2}m\,(a + a_0)$

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**Using (ii) in this** $F = \frac{3}{2}ma + \frac{1}{2}ma_0$ **...(v)**

**From (ii) and (iii)** $F = ma_0 - ma$ **...(vi)**

$\ldots.(\mathrm{vi})$

**Solving** $(\mathrm{v})$ **and (vi)** $a = \frac{F}{4m}$

**Q.3 Answer:**

$\alpha_A > \alpha_B$

**Solution:**

**In figure,** $(\mathrm{T_2 - T_1})\,\mathrm{R} = \frac{\mathrm{MR^2}}{2}\alpha_\mathrm{A}$

$\mathrm{T_1 - Mg = Ma_A}$

$\mathrm{T_2 = 2Mg}$

$\mathrm{a_A = Ra_A}$

$\alpha_\mathrm{A} = \frac{2\,\mathrm{g}}{3\mathrm{R}}$

**From figure,**

$(\mathrm{T_2 - T_1}) \times \mathrm{R} = \frac{\mathrm{MR^2}}{2}\mathrm{a_B}$

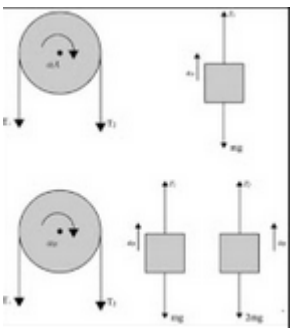
$\mathrm{T_1 - Mg = Ma_B}$

$\mathrm{2Mg - T_2 = 2Ma_B}$

$a_B = ra_B$

$\alpha_B = \frac{2g}{7R}$

**So,** $\alpha_A > \alpha_B$



**Q.4 Answer:**

$\frac{v}{L}$

**Solution:**

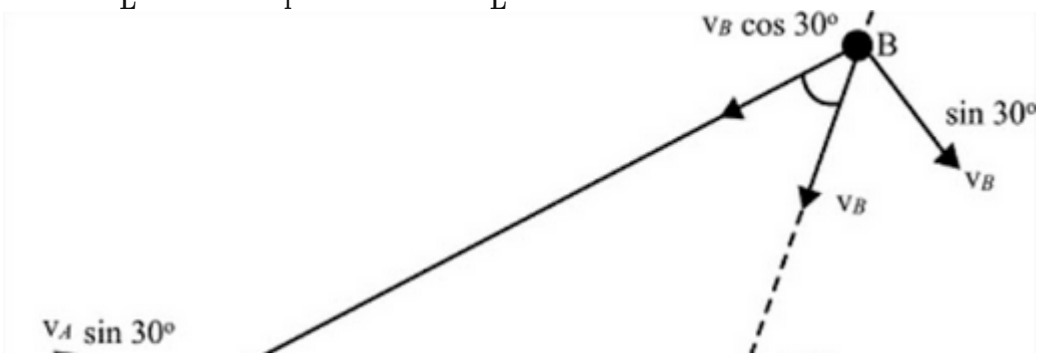
**The velocities of the ends** $A$ **and** $B$ **along the length of rod should be the same.**

**Hence,**

$\mathrm{v_A \cos 30^\circ = v_B \cos 30^\circ \Rightarrow v_A = v_B = v}$

**Hence, the angular velocity of the rod is**

$\omega = \frac{(\mathrm{v_{AB}})^\perp}{\mathrm{L}} = \frac{\mathrm{2v \sin 30^\circ}}{\mathrm{l}} \Rightarrow \omega = \frac{\mathrm{v}}{\mathrm{L}}$




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**Q.5 Answer:**

$\frac{181}{2} MR^2$

**Solution:**

$$I_0 = I_{cm} + md^2$$
$$= \frac{7MR^2}{2} + 6\left(M \times (2R)^2\right)$$
$$= \frac{55MR^2}{2}$$
$$= I_p = I_0 + md^2$$
$$= \frac{55MR^2}{2} + 7M(3R)^2$$
$$= \frac{181}{2}MR^2$$

**Q.6 Answer:**

$Mg/3$

**Solution:**

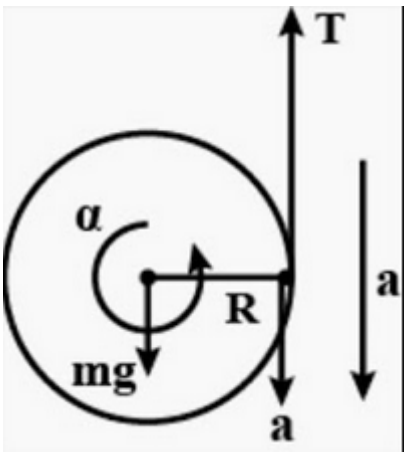
$$\tau = Ra$$
$$TR = Ia$$

**For solid cylinder** $I = \frac{1}{2}MR^2$ **and** $a = r\alpha$ **(no slipping)**

$$\Longrightarrow T = \frac{1}{2}Ma$$

**Equation of motion,** $Ma = Mg - T$

$$2T = Mg - T \Longrightarrow T = \frac{Mg}{3}$$




**Q.7 Answer:**

more than 3 but less than 6

**Solution:**

**to reverse the direction**

$$\int \tau d\theta = 0$$
$$\tau = \left(20t - 5t^2\right) \times 2 = 40t - 10t^2$$
$$\alpha = \frac{\tau}{I} = \frac{40t - 10t^2}{10} = 4t - t^2$$

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$\omega = \int_0^t a dt = 2t^2 - \frac{t^3}{3}$

$\text{at } \omega = 0 \Longrightarrow 2t^2 - \frac{t^3}{3} = 0 \Longrightarrow t^3 = 6t^2 \Longrightarrow t = 6$

$\theta = \int_0^6 \omega dt = \int_0^6 \left(2t^2 - \frac{t^3}{3}\right) dt = \left[\frac{2t^3}{3} - \frac{t^4}{12}\right]_0^6 = 36\text{rad} = \frac{36}{2\pi} < 6$

**Q.8 Answer:**

$\sqrt{(a + r\alpha)^2 + (r\omega^2)^2}$

**Solution:**

**We have two components of acceleration of point P- one in radial direction which is $r\omega^2$ and the other in tangential direction which is $ra$. Also, every point on the disc has a forward acceleration of a. Thus we have total acceleration in the forward direction as a $+r\alpha$.**

**Thus we get the resultant acceleration of point P as $\sqrt{(a + ra)^2 + (r\omega^2)^2}$.**

**Q.9 Answer:**

$\frac{M \sin\theta}{1 - \sin\theta}$

**Solution:**

**According to diag.**

$T = mg$

$f = Mg \sin\theta + T \sin\theta$

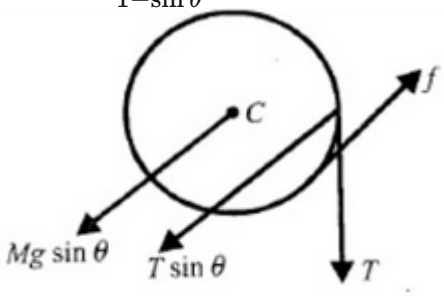
$= Mg \sin\theta + mg \sin\theta$

**Balancing torque about C;**

$Tr = fr$

$\Rightarrow mgr = (Mg \sin\theta + mg \sin\theta)r$

$\Rightarrow m = \frac{M \sin\theta}{1 - \sin\theta}$




**Q.10 Answer:**

$\frac{h}{l} = \frac{3\sqrt{3}}{16}, f = \frac{16\sqrt{3}}{3}\ \text{N}$

**Solution:**

**Balancing torque about the lowest point, we get**

$N \frac{h}{\sin 60^0} = mg \frac{1}{2} \cos 60^\circ$

**Also,** $N + \frac{N}{2} = mg$

$N = \frac{2mg}{3}$

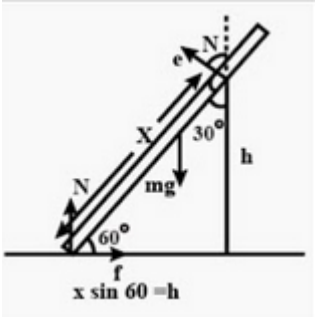
$\frac{4mg}{3\sqrt{3}} h = mg \frac{1}{4}$

$\frac{h}{1} = \frac{3\sqrt{3}}{16}$

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$f = N \sin 60^0 = \frac{mg}{\sqrt{3}} = \frac{16}{\sqrt{3}}$

**Answer is option D.**




**Q.11 Answer:**

$\frac{(M-m)g}{\left(M+m+\frac{2I}{r^2}\right)}$

**Solution:**

**Since the pulleys in the question are not massless, the tension in the string cannot be uniform. There should be different tensions in each part of the string to account for the torque required for the angular acceleration of the pulleys.**

**Let the acceleration of the boxes be $a$ and the angular acceleration of the pulleys be $\alpha$**

**Let the tension in the portion of the string attached to $\mathrm{M}$ be $\mathrm{T_1}$ Let the tension in the portion of the string between the pulleys be $\mathrm{T_2}$ Let the tension in the portion of the string attached to $\mathrm{m}$ be $\mathrm{T_3}$**

**Since the rope does not slip on the pulleys,** $\mathrm{a = r\alpha}$

**Examining the forces on** $\mathrm{M}$, $\mathrm{Mg - T_1 = Ma - (1)}$

**Examining the forces on** $\mathrm{m}$, $\mathrm{T_3 - mg = ma - (2)}$

**Examining the torque acting on the right pulley,** $\mathrm{(T_1 - T_2)\,r = \frac{Ia}{r} - (3)}$ **Examining the torque acting on the left pulley,** $\mathrm{(T_2 - T_3)\,r = \frac{Ia}{r} - (4)}$

**Adding (1) and (2),** $\mathrm{T_3 - T_1 = (M + m)a + (m - M)g - (5)}$ **Adding** $(3)$ **and (4),** $\mathrm{T_1 - T_3 = \frac{2Ia}{r^2} - (6)}$

**Adding** $(5)$ **and (6),** $a\left(\frac{2I}{r^2} + (M+m)\right) = (M-m)g$

$\Rightarrow \mathrm{a = \frac{(M-m)g}{\left(M+m+_{r^2}\right)}}$

**Q.12 Answer:**

outside the sphere

**Solution:**

**The instantaneous axis of rotation will fall on the outside of the sphere. Since in case of pure rotation, the instantaneous axis of rotation will be on the centre. In case of pure rolling, the instantaneous axis of rotation will be on P. But if** $\mathrm{v_c > \omega R}$**, then the instantaneous axis of rotation will fall out of the sphere.**

**Q.13 Answer:**

the block will remain at rest on the plane up to certain $\theta$ and then it will topple

**Solution:**

**As we increase the angle of inclination, the frictional force acts in upward direction on incline to prevent**

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**sliding of the block.**

**But this frictional force also causes torque about the CoM of the block, which is countered by the torque due to the Normal force on the block.**

**But as the angle of inclination $\theta$ increases, the normal force reduces and perpendicular distance of Normal force from CoM also reduces which decreases the counter-torque due to the normal force. So after an angle, the torque due to frictional force overcomes that due to normal force, and the block starts to topple.**

**Q.14 Answer:**

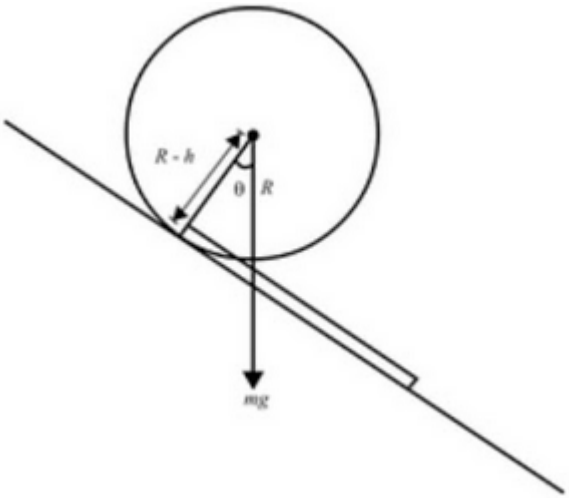
$R(1-\cos\theta)$

**Solution:**

**The sphere is on the verge of toppling when a line of action of weight passes through the edge. The equilibrium is because of the line of action of weight and the normal reaction provided by the edge.**

$\cos\theta = \frac{(\mathrm{R}-\mathrm{h})}{\mathrm{R}}$

**or** $\mathrm{h} = \mathrm{R} - \mathrm{R}\cos\theta$




**Q.15 Answer:**

$< 1$

**Solution:**

**moment of inertia along x-axis $<$ moment of inertia along y-axis. because $\mathrm{AB}$ and DE wires do not appear in moment of inertia of $x$-axis equation,but remaining remains the same.**

**Therefore, ratio of moment of inertia along x-axis to moment of inertia along $y$-axis is $< 1$.**

**Q.16 Answer:**

$5\mathrm{Mr}^2$

**Solution:**

**The moment of inertia of the disc about the centroidal axis is $\frac{\mathrm{Mr}^2}{2}$. The moment of inertia of the disc at A about the axis passing through B is given using parallel axis theorem as $\frac{\mathrm{Mr}^2}{2} + \mathrm{M}(2\mathrm{R})^2$**

**Thus the total moment of inertia of the whole system is given as $\frac{\mathrm{Mr}^2}{2} + \frac{\mathrm{Mr}^2}{2} + \mathrm{M}(2\mathrm{R})^2 = 5\mathrm{Mr}^2$**

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**Q.17 Answer:**

$\frac{2}{3}mg$

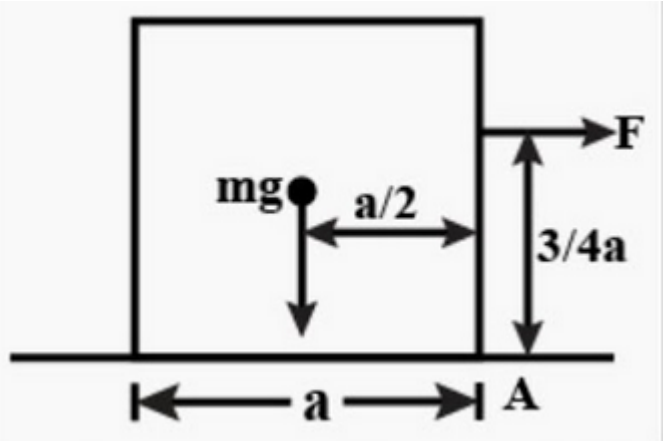
**Solution:**

**The minimum force at which the cube will tilt will be when the torques of the forces acting about A will be equal.**

**Thus we get** $\mathrm{mg}\left(\frac{\mathrm{a}}{2}\right) = \mathrm{F}\left(\frac{3}{4}\mathrm{a}\right)$

**Thus we get**

$\mathrm{F} = \frac{2}{3}\mathrm{mg}$




**Q.18 Answer:**

$a_{\mathrm{cm}} = 2\alpha r$

**Solution:**

**When the lower disc has an angluar acceleration of a the upper disc also accelerates at a, they being identical and the tension on rope is same for both discs**

$\alpha = \frac{\partial\omega}{\partial \mathrm{t}}$

$\mathrm{a_{cm}} = \frac{\partial \mathrm{r}\omega}{\partial \mathrm{t}} + \frac{\partial \mathrm{r}\omega}{\partial \mathrm{t}} = 2\mathrm{r}\alpha$

**Q.19 Answer:**

$\vec{v}_{\bar{F}} = \left(v + \frac{vr\sin\theta}{R}\right)\hat{i} - \frac{vr\cos\theta}{R}\hat{j}$

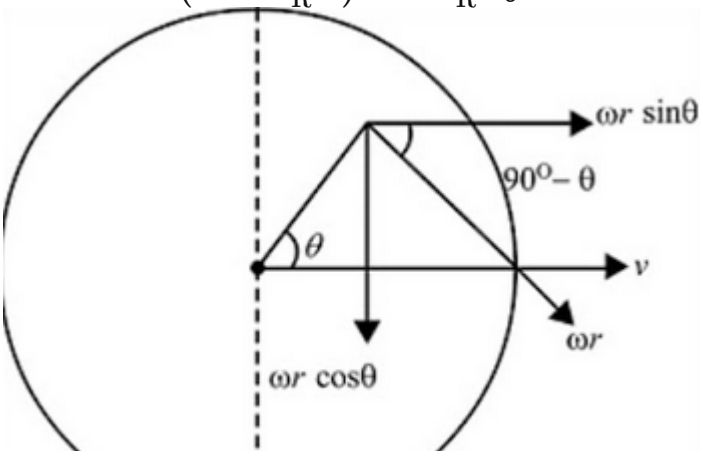
**Solution:**

**Here,** $\omega = \frac{\mathrm{v}}{\mathrm{R}}$

$\mathrm{V_{P_x}} = \left(\mathrm{v} + \frac{\mathrm{v}}{\mathrm{R}}\mathrm{r}\sin\theta\right)\hat{\mathrm{i}}$

$\mathrm{V_{P_y}} = -\left(\frac{\mathrm{v}}{\mathrm{R}}\mathrm{r}\cos\theta\right)\hat{\mathrm{j}}$

$\therefore \overrightarrow{\mathrm{V_P}} = \left(\mathrm{v} + \frac{\mathrm{vr}\sin\theta}{\mathrm{R}}\right)\hat{\mathrm{i}} - \frac{\mathrm{vr}\cos\theta}{\mathrm{R}}\hat{\mathrm{j}}$




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**Q.20 Answer:**

North

**Solution:**

**By using Thumb rule the angular velocity in north direction. Hence, the answer is North.**

**Q.21 Answer:**

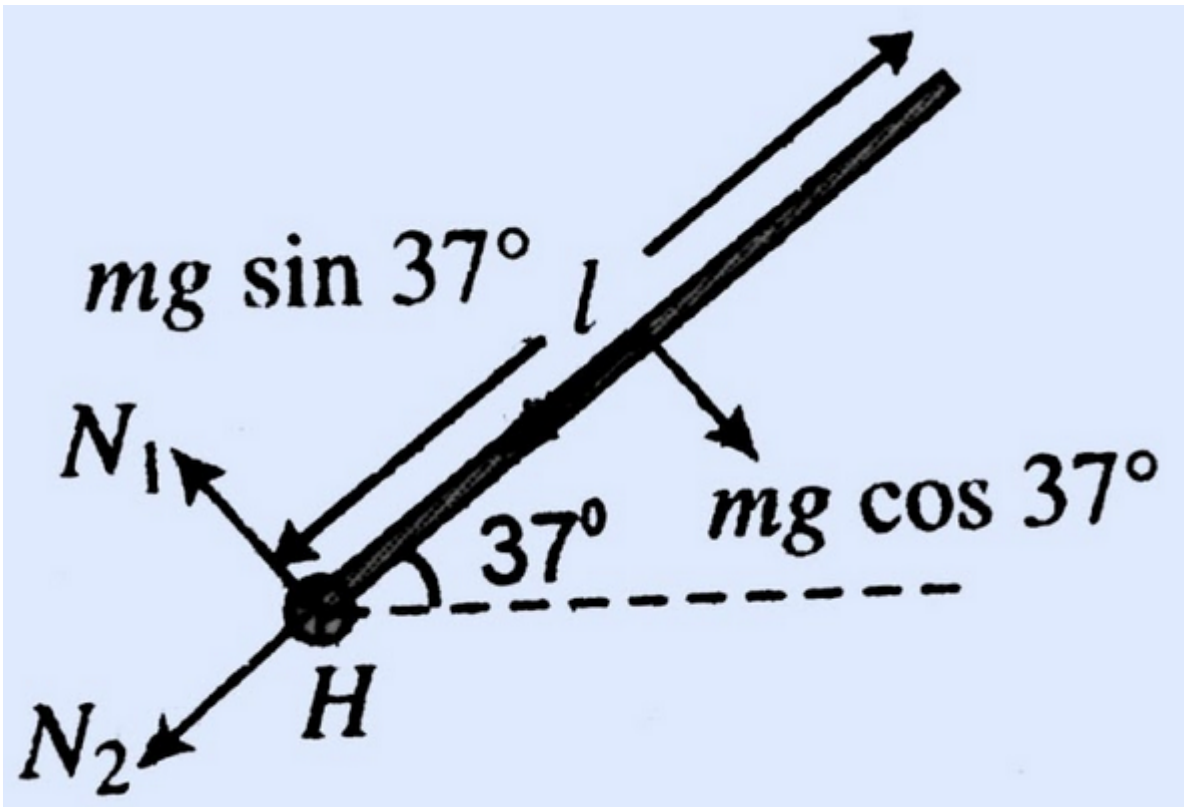
—

**Solution:**

**Torque about hing** $= \tau H = I\alpha$

$Mg\frac{\cos 37^\circ l}{2} = \frac{ml^2}{3}\alpha$

$\alpha 6g/5l$




$a_t = \alpha\frac{l}{2} = \frac{3g}{5}$

$Mg\cos 37^\circ - N - 1 = ma_1$

$N_1 = \frac{mg}{5}$

**Angular velocity of rod is zero so** $N - 2 = mg\sin 37^\circ = 3mg/5$

$N = \sqrt{N - 1^2 + N_2^2} = \sqrt{\left(\frac{mg}{5}\right)^2 + \left(\frac{3mg}{5}\right)^2} = \frac{mg\sqrt{10}}{5}$

**Q.22 Answer:**

—

**Solution:**

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**The loss in gravitional potential energy of the load = Gain in kinetic energy of** b

$\Rightarrow \mathrm{mgl}\left(1-\sin 30^{0}\right)=\frac{1}{2} \mathrm{Iw}_{2}+\frac{1}{2} \mathrm{Mu}^{2}$

$\Rightarrow \frac{\mathrm{mgl}}{2}=\frac{1}{2} \mathrm{ml}^{2} \mathrm{w}^{2}+\frac{1}{2} \mathrm{Mu}^{2}$

**Since the load and body have beenin contact upto this point,** $1 \mathrm{w} \sin 30^{\circ}=\mathrm{u}$ **or,** $l \mathrm{w}=2 \mathrm{u}$

**therefore the above equation becomes,**

$\frac{\mathrm{mgl}}{2}=2 \mathrm{mu}^{2}+\frac{1}{2} \mathrm{Mu}^{2}$

**Now consider the free body diagram of the load.**

**Centreipital acceleration is equal to the component of weight along the length of sin ce the body has just lost the contact.**

$\Rightarrow \mathrm{mg} \sin 30^{\circ}=\mathrm{mw}^{2} \mathrm{r}$

$\Rightarrow \mathrm{w}^{2}=\frac{\mathrm{g}}{2 \mathrm{l}}$

$\Rightarrow \mathrm{u}^{2}=\frac{\mathrm{g} l}{8}=\frac{1}{2} \sqrt{\frac{\mathrm{gl}}{2}}$

$\mathrm{u} \sin \mathrm{g}$ **this in above equation gives,**

$\frac{\mathrm{M}}{\mathrm{m}}=4$

**Q.23 Answer:**

—

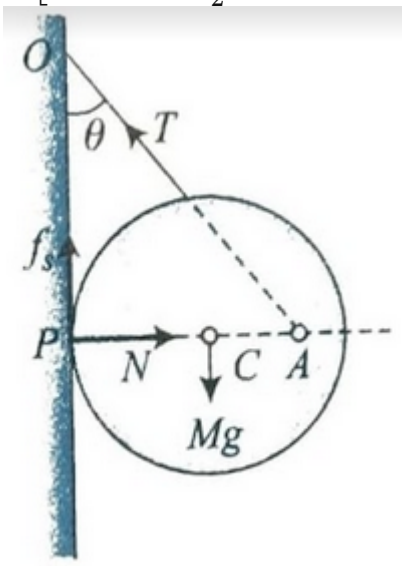
**Solution:**

**Net torque about** $A=0$

$f_s \frac{3R}{2}=Mg \cdot \frac{R}{2}$

$\therefore f_s=\frac{Mg}{3}$

$\left[\therefore PA=\frac{3R}{2} \tan 45^{\circ}=\frac{3R}{2}\right] \quad \ldots(\mathrm{i})$




**Horizontal equilibrium:** $N=\frac{T}{\sqrt{2}}$ **...(ii)**

**Vertical equilibrium:** $\frac{T}{\sqrt{2}}+f_s=Mg$

$T=\frac{2\sqrt{2}}{3} Mg$ **...(iii)**

**Put in (ii)** $N=\frac{2}{3} Mg$

$f_s \leq \mu_s N \Rightarrow \frac{Mg}{3} \leq \mu_s \cdot \frac{2}{3} Mg \Rightarrow \frac{1}{2} \leq \mu_s$

**Hence,** $\mu_{\min }=0.50$

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**Q.24 Answer:**

—

**Solution:**

**Let the centre of mass of the block move down with an acceleration a.** $\mathrm{mg}-\mathrm{T}=\mathrm{ma}$

**Let the block have an angular acceleration a about its centre of mass. The point A has only horizontal acceleration at the instant at which the string** $\mathrm{Q}$ **is cut.**

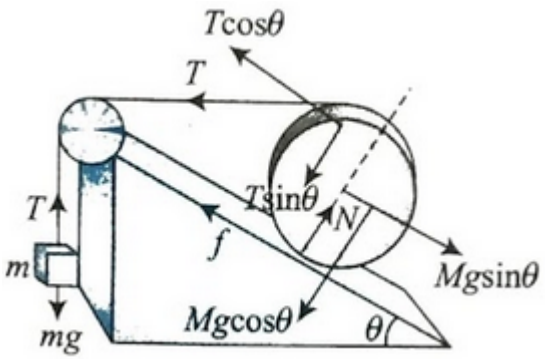
$$\mathrm{T}\frac{1}{2}=\frac{\mathrm{ml}^2}{6}\mathrm{a}$$
$$a\frac{1}{\sqrt{2}}\cos 45=\mathrm{a}$$
$$\mathrm{a}=6\ \mathrm{ms}^{-2}$$

**Q.25 Answer:**

—

**Solution:**

**Let the mass of the block be** $m$




**For equilibrium of the block,** $T=mg$ **...(i)**

**For rotational equilibrium of the cylinder, net torque about its centre must be zero.**

$$\therefore \quad TR=fR \quad [\mathrm{R}=\text{ radius of cylinder}]$$
$$\Rightarrow mg=f \quad \ldots(\mathrm{ii})$$

**For translational equilibrium of the cylinder**

$$N=Mg\cos\theta+T\sin\theta$$
$$\Rightarrow \quad N=Mg\cos\theta+mg\sin\theta \quad \ldots(\mathrm{iii})$$

**and** $f+T\cos\theta=Mg\sin\theta$ **...(iv)**

$$\Rightarrow f(1+\cos\theta)=Mg\sin\theta$$
$$\Rightarrow f=\frac{Mg\sin\theta}{(1+\cos\theta)} \Rightarrow mg=\frac{Mg\sin\theta}{1+\cos\theta}$$
$$m=\frac{M\sin\theta}{1+\cos\theta} \quad \ldots(\mathrm{v})$$

**On substituting the values of** $M$ **and** $\theta$ **we get** $m=30\ \mathrm{kg}$

**Q.26 Answer:**

—

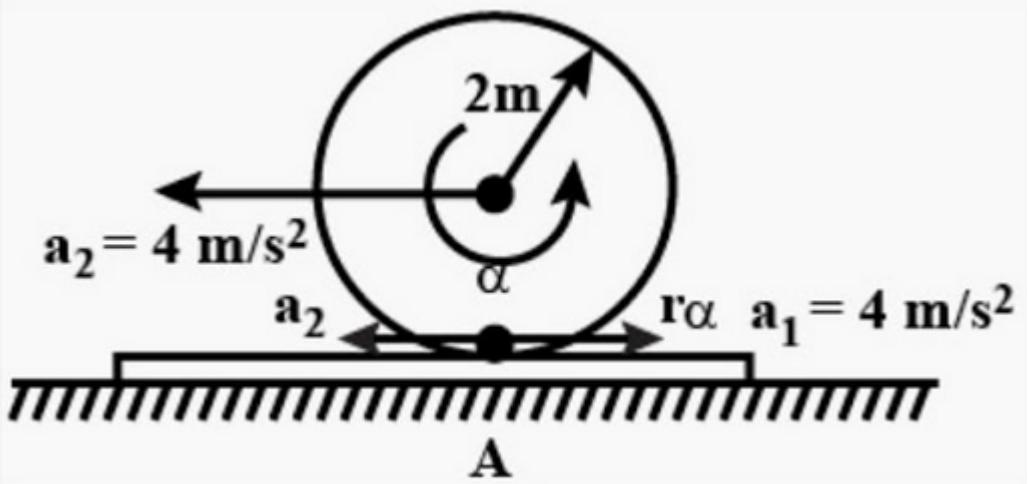
**Solution:**

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**Acceleration of point A must be equal to the acceleration of the plank due to pure rolling.**

$a_r - a_2 = a_1$

$a(2) - 4 = 2 \Longrightarrow a = 3\text{rad/s}^2$




**Q.27 Answer:**

—

**Solution:**

**The rod will rotate about the end A.**

**Let a be the linear acceleration of** $\mathrm{COM}$ **of rod and** $\alpha$ **be the angular acceleration.**

$$a = \frac{\tau}{\mathrm{I}} = \frac{\mathrm{mg}\frac{1}{2}}{\frac{\mathrm{ml}^2}{3}} = \frac{3\,\mathrm{g}}{2\mathrm{l}}$$

$$\mathrm{a} = \frac{1}{2}\alpha = \frac{1}{2}\frac{3\,\mathrm{g}}{2\mathrm{l}} = \frac{3\,\mathrm{g}}{4}$$

**From force equation:**

$$\mathrm{mg} - \mathrm{F} = \mathrm{ma}$$

$$\mathrm{F} = \mathrm{mg} - \mathrm{m}\frac{3\,\mathrm{g}}{4} = \mathrm{m}\frac{\mathrm{g}}{4} = \frac{20}{4} = 5$$

**Q.28 Answer:**

—

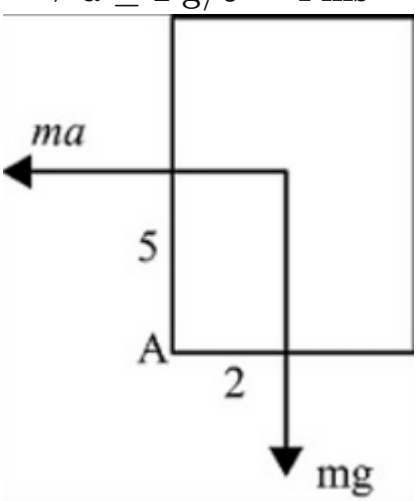
**Solution:**

**Maximum acceleration of cart so that cylinder does not slip:** $\mathrm{a_m} = \mu\mathrm{g} = 0.5 \times 10 = 5\ \mathrm{ms}^{-2}$

**For tipping over: Let acceleration of cart is a. Considering torque about A:**

$$\mathrm{Ma} \times 5 \geq \mathrm{mg} \times 2$$

$$\Rightarrow \mathrm{a} \geq 2\,\mathrm{g}/5 = 4\ \mathrm{ms}^{-2}$$




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**Q.29 Answer:**

—

**Solution:**

**Let us assume the lengths to be $z$ and $y$.**

**$\therefore$ Total length $l = y + z$.**

**Assuming the distance in the cower to be**

$$l = y + \sqrt{y^2 + l_1^2}$$

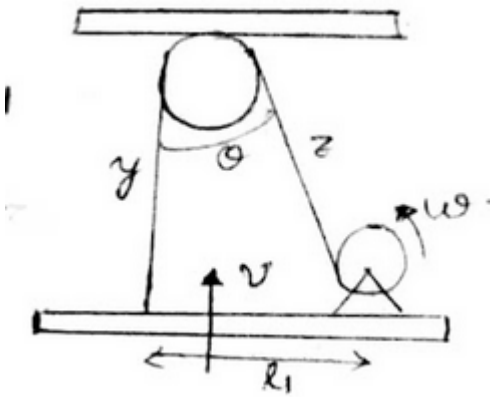
$$\Rightarrow \frac{dl}{dt} = \frac{dy}{dt} + \frac{2y}{x}\frac{dy}{\sqrt{y^2 + l_1}}\frac{dy}{dt}$$

$$\text{Here } \frac{dy}{dt} = v$$

$$\omega R = v + \cos\theta\nu$$

$$\omega_R = v(1 + \cos\theta)$$

$$v = \frac{\omega R}{1 + \cos\theta}$$




**$\therefore$ (b) $\frac{\omega R}{1+\cos\theta}$ is the correct option.**

**Q.30 Answer:**

—

**Solution:**

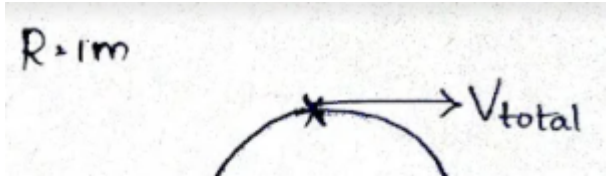
**Rolling:**

*** No Relative motion**

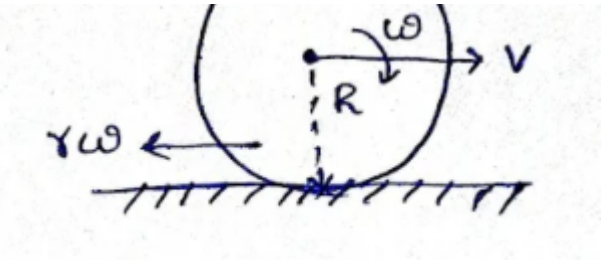
$$V = R\omega$$

$$V_{\text{total}} = v + R\omega$$

$$= v + v = 2\ \text{V}$$




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**het 'r be the required radius**

**- Centripetal**

**nen** $F_{\text{orce}} = \frac{(V_{\text{total}})^2}{\gamma} = \frac{v^2}{R}$

$\Rightarrow \frac{\left(2v'\right)^2}{r} = \frac{V^2}{R}$

$r = 4R$

$\Rightarrow r = 4m$

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